

Due: Thursday, November 30th at the beginning of class.

Read Section 2.5 and 3.2 of *Levin*.

From *Levin*, complete the following:

Section 3.2: # 3, 5, 8, 16

Section 2.5: # 3, 6, 10

1. We say that a divides b , written $a|b$, if there is some integer k such that $ak = b$.
 - (a) Show that $3|12$ with the definition.
 - (b) Show that $3|9$ with the definition.
 - (c) What can you say about $3|(12 + 9)$, using the previous two answers.
 - (d) Show that $3|0$ with the definition.
 - (e) Prove that if $a|b$ and $a|c$, then $a|(b + c)$.
2. Prove that $\sqrt{7}$ is irrational. (You may assume that if n^2 is a multiple of 7, then n is a multiple of 7.)
3. Prove that $8 + 11 + 14 + \cdots + (3n + 5) = \frac{3n^2 + 13n}{2}$, for $n = 1, 2, 3, \dots$ via induction.
4. Prove that $3|4^n - 1$ for $n = 0, 1, 2, \dots$ via induction.